SMART AGRICULTURE EXPERT SYSTEM

A Synopsis

for

Minor Project

BACHELOR OF TECHNOLOGY

in

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BY

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1. Introduction

Agriculture continues to be a foundation of the Indian economy, with over fifty-eight percent of rural households depending directly on farming for their primary livelihood (Ministry of Agriculture & Farmers Welfare, Government of India). Despite this significance, a considerable proportion of farming communities still follow conventional practices when evaluating field health, deciding on pest management strategies, and scheduling irrigation cycles. These choices tend to emerge from accumulated generational experience rather than from structured analytical reasoning. Such reliance on subjective observation often leads to delayed interventions, unnecessary chemical usage, avoidable crop losses, and lowered seasonal yields.

Parallel developments in computational intelligence — spanning supervised classification, deep convolutional feature extraction, and real-time environmental data retrieval — have created practical openings for building intelligent advisory platforms tailored to agricultural contexts. Systems that combine structured domain knowledge with statistical pattern recognition can bridge the gap between expert agronomic judgment and data-informed field management.

The present work introduces a hybrid intelligent agricultural advisory system that integrates three complementary reasoning pathways: a deterministic heuristic rule engine modelled on established agronomic safety constraints, a calibrated gradient-boosted tabular classifier trained on structured field observation logs, and a multi-backbone convolutional neural network ensemble trained on labelled leaf imagery for visual disease identification. These three pathways converge through a weighted confidence fusion mechanism that produces a single, interpretable recommendation for the farmer.

Beyond static analysis, the system incorporates a live environmental intelligence layer that retrieves real-time meteorological measurements — temperature, wind velocity, and precipitation status — from public geocoding and forecast application programming interfaces, thereby anchoring its recommendations in current atmospheric conditions rather than farmer-reported approximations.

The system archives every consultation into a persistent relational data store, enabling longitudinal trend analysis and outbreak alerting across repeated visits. A structured digital advisory report — containing the recommended action, supporting rationale, confidence metrics, chemical and organic treatment prescriptions, spray timing windows, and follow-up schedules — is generated for each session and exported as a portable text document.

2. Literature Review

The intersection of computational methods and agricultural decision-making has been the subject of growing scholarly attention over the past decade.

Machine Learning for Agricultural Prediction

Supervised learning algorithms — including Random Forest, Support Vector Machines, and gradient-boosted decision trees — have demonstrated reliable performance in structured agricultural classification tasks such as crop yield estimation, irrigation scheduling, and pest outbreak prediction. Kamilaris and Prenafeta-Boldu (2018) presented a comprehensive survey highlighting the applicability of deep learning across multiple agricultural domains, noting that ensemble methods and gradient boosting approaches frequently outperform single-model baselines when the input feature space is heterogeneous and moderately sized [1]. XGBoost, in particular, has shown strong results in tabular agricultural prediction tasks due to its regularization properties and handling of missing values (Chen and Guestrin, 2016) [12].

Convolutional Neural Networks for Plant Disease Detection

The visual identification of plant diseases through leaf image analysis has been significantly advanced by deep convolutional architectures. Mohanty, Hughes, and Salathé (2016) demonstrated that convolutional neural networks trained on the PlantVillage benchmark dataset could distinguish between multiple disease categories with classification accuracy exceeding ninety percent under controlled imaging conditions [2]. Subsequent work by Ferentinos (2018) extended these results across broader crop species and disease combinations, confirming the generalizability of transfer learning from ImageNet-pretrained feature extractors [7]. Sladojevic et al. (2016) further validated the viability of deep neural

network architectures for automated leaf disease recognition, achieving strong per-class precision metrics [5]. Recent ensemble strategies — combining multiple pretrained backbones such as EfficientNet, ResNet, and MobileNet through logit-level or probability-level fusion — have been shown to reduce variance and improve robustness relative to single-architecture baselines.

Expert Systems and Rule-Based Agricultural Advisors

Knowledge-based expert systems that encode domain-specific agronomic rules have a long history in agricultural informatics. Zhang and Kovacs (2012) discussed the integration of sensor-based field observations with rule-driven reasoning engines for precision agriculture applications [3]. While purely rule-based systems provide transparent and interpretable outputs, they may lack the statistical adaptability needed when environmental conditions deviate from the patterns encoded in their rule sets.

Hybrid Intelligent Systems

Wolfert et al. (2017) reviewed the convergence of big data analytics, machine learning, and domain expertise within smart farming platforms, arguing that hybrid architectures combining structured expert knowledge with data-driven statistical models yield more reliable recommendations than either approach in isolation [4]. The present work aligns with this hybrid paradigm by fusing heuristic safety constraints with calibrated probabilistic ML outputs through a weighted confidence index.

Probability Calibration

In decision-critical agricultural contexts, it is essential that a classifier's predicted probability faithfully reflects the true likelihood of the predicted outcome. Isotonic regression-based probability calibration, as implemented through the Platt scaling and isotonic calibration methods available in scikit-learn's CalibratedClassifierCV, has been shown to improve the reliability of confidence estimates from boosted tree classifiers (Niculescu-Mizil and Caruana, 2005) [13].

Real-Time Environmental Data Integration

Modern weather forecast APIs — including Open-Meteo and OpenWeatherMap — provide programmatic access to current atmospheric observations with global geographic coverage.

Integrating live environmental data into agricultural advisory pipelines enables context-aware recommendations that account for real-time wind drift risk, precipitation interference, and temperature suitability for chemical application.

3. Problem Definition

Field-level crop management decisions — including the timing, method, and intensity of pest control interventions — depend on the simultaneous evaluation of multiple interacting environmental and biological variables. Soil moisture content, standing water availability, insect population density, leaf surface condition, ambient wind intensity, precipitation probability, crop growth stage, and cultivated area all influence whether a chemical or organic intervention is warranted, safe, and likely to be effective.

Manual evaluation of these parameters is inherently limited by subjective variability, incomplete observation, and the absence of quantitative benchmarks. Spraying chemical pesticides during high-wind conditions may result in hazardous aerosol drift beyond the target area. Applying treatments immediately before anticipated rainfall diminishes active-ingredient retention on leaf surfaces. Blanket spraying across an entire field when only localized damage is present wastes resources and increases soil contamination. Conversely, failing to act promptly when disease symptoms are visually apparent can allow pathogen spread beyond recoverable thresholds.

Existing agricultural advisory tools typically adopt one of two isolated approaches: static rule-based systems that apply predetermined thresholds to individual parameters without learning from accumulated data, or purely statistical machine learning models that may produce accurate predictions but lack the interpretive transparency and domain safety constraints that farmers and extension officers require.

There is therefore a demonstrated need for an integrated intelligent system that can simultaneously analyze multiple field parameters through complementary analytical pathways — combining transparent rule-based safety evaluation with calibrated probabilistic machine learning classification and optional visual disease identification — and synthesize

these perspectives into a single, confidence-weighted recommendation that the farmer can act upon with an informed understanding of the supporting rationale.

4. Objectives

The principal objectives of the Smart Agriculture Expert System are enumerated below:

1. To architect and implement a modular, production-grade hybrid intelligent system for agricultural decision support, following object-oriented software engineering principles with centralized configuration management, structured logging, and persistent data archival.

2. To design a multi-parameter field observation intake pipeline capable of capturing nine categorical agricultural variables: soil condition, field water availability, insect presence intensity, leaf surface condition, wind intensity, weather status, crop type, crop growth stage, and cultivated field size.

3. To construct a deterministic rule-based expert engine that evaluates field observations against established agronomic safety constraints — including wind drift risk thresholds and heavy rain chemical washout guards — and produces an interpretable heuristic confidence score.

4. To train and calibrate a gradient-boosted tabular classification model (XGBoost) on structured agricultural observation logs, employing Stratified K-Fold cross-validation for stability assessment and Isotonic regression for probability calibration.

5. To develop a multi-architecture convolutional neural network ensemble — comprising EfficientNet-B0, ResNet-18, and MobileNetV3 backbones with weighted logit-level fusion — for visual leaf disease classification across a standardized thirty-eight-class plant disease taxonomy.

6. To implement a weighted hybrid decision fusion mechanism (Custom Confidence Index) that synthesizes rule-engine heuristic confidence and calibrated ML probability into a unified decision score.

7. To integrate a live environmental intelligence service that retrieves real-time temperature, wind speed, and sky condition data from public geocoding and weather forecast APIs, eliminating dependence on farmer-reported weather approximations.

8. To generate structured farmer advisory outputs — including recommended action labels, chemical and organic pesticide prescriptions with dosage guidelines, spray timing windows, follow-up schedules, and technical diagnostic reports — and to export these as persistent digital receipt documents.

9. To maintain a relational consultation archive with indexed spatial-temporal queries, enabling longitudinal trend analysis and repeated-disease outbreak alerting.

5. System Architecture

The Smart Agriculture Expert System is structured as a modular pipeline architecture consisting of interconnected subsystems. The following diagram illustrates the high-level data flow across the system components:

5.1 System Data Flow

The system operates through a six-stage pipeline: (1) Real-time weather satellite synchronization via Open-Meteo and OpenWeatherMap REST APIs; (2) Multi-parameter field observation collection through an interactive CLI interface; (3) Optional leaf image analysis through a triple-CNN vision ensemble; (4) Parallel evaluation through the rule-based expert engine and calibrated XGBoost ML predictor; (5) Hybrid confidence fusion via the Custom Confidence Index (CCI) weighted mechanism; and (6) Advisory report generation, database archival, and digital prescription export.

5.2 Module Inventory

Module	Source File	Responsibility
Centralized Configuration	config.py	Dataclass-based system paths, API keys, model parameters, fusion thresholds, pesticide knowledge base
Feature Mapping	mapper.py	Categorical-to-numeric encoding for ML-

		compatible feature representation
Rule-Based Expert Engine	decision_engine.py	Deterministic heuristic evaluation with wind drift and rain safety guardrails
Tabular ML Predictor	predict_action.py	Calibrated XGBoost classification with feature alignment and probability output
Vision Ensemble	ensemble_predict.py	Multi-backbone PyTorch CNN ensemble (EfficientNet-B0 + ResNet-18 + MobileNetV3)
Weather Intelligence	run_agri_sense.py	Real-time geocoding and meteorological data retrieval via REST APIs
Hybrid Fusion Engine	run_agri_sense.py	Weighted CCI computation combining rule confidence and ML probability
Advisory Reporter	run_agri_sense.py	Farmer advisory card rendering, technical report, digital receipt export
Database Manager	database_manager.py	SQLite schema management, transactional logging, indexed historical queries
Synthetic Data Generator	generate_synthetic_data.py	Randomized agricultural observation log generation for model training
Training Pipelines	train_tabular.py, train_cv.py, calibrate.py	XGBoost training, 5-Fold stratified CV, Isotonic calibration
CNN Training Pipelines	train_pest_cnn.py, train_pest_cv.py	Keras CNN and EfficientNet-based pest model training
Consultation Log Viewer	view_logs.py	Terminal dashboard for historical consultation analysis
Application Entry Point	run_agri_sense.py	CLI argument parsing, pipeline orchestration, end-to-end execution

6. Methodology

6.1 Data Collection and Synthetic Augmentation

The system operates on structured field observation records comprising nine categorical agricultural parameters: soil moisture condition (Wet / Normal / Dry / Hard-Dry), field water availability (Too much / Normal / Low / Very low), insect presence intensity (Many / Few / None / Not sure), leaf surface condition (Holes-eaten / Spots-color change / Curled leaves / Leaves normal), wind intensity (Strong / Light / No wind), weather status (Clear / Cloudy / Light rain / Heavy rain / Not sure), crop type (Maize / Wheat / Rice / Cotton / Vegetables / Other), crop growth stage (Early / Mid / Late), and cultivated field size (Small / Medium / Large).

For model training and validation, a synthetic data generation module produces realistic observation samples by randomly sampling from the defined categorical option space, computing rule-engine decisions for each generated sample, and appending the resulting records to a persistent CSV log file. This bootstrap strategy ensures sufficient sample diversity for robust classifier training even in the absence of large-scale field-collected datasets.

For the visual disease detection pathway, leaf imagery is drawn from the publicly available PlantVillage benchmark dataset, which contains labelled images spanning thirty-eight disease and healthy-plant categories across fourteen crop species (Mohanty et al., 2016) [2].

6.2 Feature Preprocessing and Encoding

Raw categorical field observations are transformed into quantitative numerical representations through a deterministic mapping module. Each categorical level is assigned a domain-informed numeric value: for instance, soil moisture categories map to estimated volumetric moisture percentages (Wet → 47.0, Normal → 35.0, Dry → 22.0, Hard-Dry → 10.0), while insect presence maps to estimated count densities (Many → 20.0, Few → 7.0, None → 0.0). Crop type is encoded via one-hot dummy variable expansion to preserve nominal category structure. This combined numeric representation is aligned against the trained model's expected feature column ordering before inference.

6.3 Rule-Based Expert Engine

A deterministic heuristic evaluation engine assesses field observations against structured agronomic safety rules. The engine computes a cumulative risk score by evaluating insect activity levels (contributing 2.0 points if insects are present), leaf damage indicators (contributing 2.0 points if abnormal leaf conditions are observed), soil moisture deficits (contributing 1.0 point if soil is dry), and water availability shortfalls (contributing 1.0 point if field water is low).

Two absolute safety guardrails operate as early-exit constraints: if heavy rainfall is detected or reported, the engine immediately returns a "No Spray - Monitor" recommendation with minimal confidence (0.10), recognizing that chemical application would be washed away. Similarly, if wind intensity is classified as strong, the engine returns a monitoring recommendation with reduced confidence (0.20) to prevent hazardous spray drift.

Based on the cumulative risk score, the engine assigns one of three action labels — "Spray (patch only)" for scores at or above 4.0, "Monitor & Spot-Treat" for scores between 2.0 and 4.0, and "No Spray - Monitor" for scores below 2.0 — along with a corresponding heuristic confidence value.

6.4 Calibrated Machine Learning Classification

A gradient-boosted decision tree classifier (XGBoost, Chen and Guestrin, 2016 [12]) is trained on the structured observation log dataset to predict crop management action labels. The model configuration employs 200 estimators with a maximum tree depth of four and a learning rate of 0.1, using multiclass logarithmic loss as the optimization objective.

Model stability is assessed through Stratified 5-Fold Cross-Validation, measuring weighted F1-score consistency across partitions. Following base model training, Isotonic regression-based probability calibration is applied using scikit-learn's CalibratedClassifierCV with 5-fold internal cross-validation. This calibration step adjusts the raw predicted probabilities to better reflect true class frequencies, producing more reliable confidence estimates for downstream fusion. Calibration quality is evaluated using a multiclass Brier score and visualized through reliability diagrams plotting predicted confidence against observed accuracy.

When the XGBoost library is unavailable in the runtime environment, the system gracefully falls back to scikit-learn's HistGradientBoostingClassifier, ensuring operational continuity across deployment configurations.

6.5 Multi-Architecture Convolutional Neural Network Ensemble

The visual disease detection subsystem employs a triple-backbone ensemble architecture implemented in PyTorch and the Timm model library. Three pretrained ImageNet feature extractors — EfficientNet-B0, ResNet-18, and MobileNetV3-Large — are independently fine-tuned on the thirty-eight-class PlantVillage leaf disease taxonomy. At inference time, input leaf images are resized to 224×224 pixels, normalized using ImageNet channel statistics (mean = [0.485, 0.456, 0.406], standard deviation = [0.229, 0.224, 0.225]), and passed through all three backbone networks simultaneously.

The resulting logit vectors are combined through weighted averaging — 40% EfficientNet-B0, 30% ResNet-18, 30% MobileNetV3 — and converted to probability distributions via softmax normalization. The class with the highest fused probability is selected as the predicted disease category. This ensemble strategy reduces individual model variance and leverages the complementary feature extraction characteristics of each architecture family.

The ensemble checkpoint file stores individual state dictionaries for all three backbones, enabling atomic checkpoint loading and deterministic reproducibility. When PyTorch dependencies are absent from the deployment environment, the vision layer degrades gracefully to a skip mode, allowing the remaining analytical pathways to operate independently.

6.6 Hybrid Decision Fusion — Custom Confidence Index (CCI)

The outputs from the rule-based expert engine and the calibrated tabular ML predictor are synthesized through a weighted fusion mechanism termed the Custom Confidence Index (CCI). The CCI is computed as:

$$CCI = (0.40 \times Rule_Confidence) + (0.60 \times ML_Probability)$$

This weighting assigns greater influence to the data-driven calibrated ML component while preserving the transparent safety constraints encoded in the rule engine. The final action label

is selected as the ML prediction when CCI exceeds a configurable decision threshold (default 0.55), and defaults to the rule-engine recommendation otherwise. This design ensures that when the ML model exhibits high calibrated confidence, its prediction guides the final output, but when statistical confidence is low, the deterministic rule-engine assessment prevails.

When the optional vision ensemble detects a high-confidence disease match (confidence ≥ 0.80), the system overrides the leaf condition parameter to "Holes / eaten" before rule and ML evaluation, ensuring that strong visual evidence of disease propagates through both analytical pathways.

6.7 Real-Time Environmental Intelligence Service

The system incorporates a multi-tiered real-time weather retrieval service that fetches current atmospheric conditions based on the farmer's reported location name. The primary retrieval engine uses the Open-Meteo Geocoding API to resolve city or village names to geographic coordinates, then queries the Open-Meteo Forecast API for current temperature, wind speed, and WMO weather condition codes. Weather codes are mapped to human-readable status labels: code 0 indicates clear skies, codes 1–3 and 45–48 indicate cloudy conditions, codes 51–63 indicate light rain, and codes 65 and above indicate heavy rain. Wind speed is converted from kilometers per hour to meters per second for standardized assessment.

If the primary Open-Meteo service is unreachable, the system falls back to the OpenWeatherMap REST API as a secondary data source. If both services are unavailable, the system transitions to manual input mode, prompting the farmer to select wind and weather conditions from structured menus. This multi-tiered degradation strategy ensures operational continuity regardless of network connectivity.

When live weather data is successfully retrieved, the system automatically configures wind intensity and weather status parameters, bypassing the corresponding manual input prompts and ensuring that recommendations are anchored in measured atmospheric conditions.

6.8 Persistent Data Archival and Trend Intelligence

Every consultation session is archived in a SQLite relational database with a structured schema capturing field identifier, timestamp, farmer name, location, crop type, detected

disease label, AI confidence score, final recommended action, CCI score, weather status, and wind condition. The database schema includes composite indexes on (farmer_name, location) and (timestamp DESC) to support efficient historical retrieval and spatial-temporal querying.

A trend intelligence module queries the most recent consultation records for a given farmer and location combination. If two or more recent records indicate active disease detection, the system generates a proactive alert notifying the farmer of potential recurring pest activity, supporting early intervention and outbreak containment.

6.9 Advisory Report and Digital Prescription Generation

The system produces three structured output artifacts for each consultation:

Farmer Advisory Card: A terminal-rendered summary containing the date, time, field identifier, detected disease category and AI confidence (if applicable), crop and field condition summary, recommended action with plain-language instructions, CCI confidence percentage, optimal spray timing window, and next recommended field inspection schedule. The spray window is dynamically blocked if heavy rain or strong wind conditions are detected.

Expert Prescription: A structured chemical and organic treatment recommendation derived from an internal pesticide knowledge base indexed by disease category. Each prescription specifies a recommended chemical agent with concentration dosage, an organic alternative treatment, and safety precautions.

Technical Diagnostic Report: A research-grade audit trace presenting individual component outputs — vision ensemble prediction with probability, rule engine decision with heuristic confidence and contributing reasons, calibrated ML prediction with probability, and hybrid fusion final label with CCI score — enabling transparent post-hoc review of the system's reasoning process.

All three outputs are additionally consolidated into a portable text-format digital receipt exported to a designated reports directory with a standardized filename incorporating the farmer name, location, date, and timestamp.

7. Implementation Details

7.1 Technology Stack

Layer	Technology	Version / Specification
Programming Language	Python	3.12+
Gradient Boosting	XGBoost	3.4+
Deep Learning Framework	PyTorch	2.x (CPU/CUDA)
Model Architecture Library	Timm (PyTorch Image Models)	1.0+
Alternative DL Framework	TensorFlow / Keras	2.x
Scientific Computing	NumPy, Pandas	Standard
Machine Learning Toolkit	Scikit-Learn	1.7+
Probability Calibration	CalibratedClassifierCV	Isotonic, 5-Fold
Image Processing	Pillow (PIL), TorchVision	Standard
Visualization	Matplotlib	3.x
Database	SQLite3	Built-in (Python stdlib)
HTTP Client	Requests	Standard
Model Serialization	Joblib	Standard
Configuration	Python Dataclasses	Frozen, Centralized
Weather APIs	Open-Meteo (Primary), OpenWeatherMap (Fallback)	REST / JSON
CLI Framework	argparse	Standard
Logging	Python logging module	File + Console handlers

7.2 Project Directory Structure

```
agri-sense-x/
  src/                              Source modules
    run_agri_sense.py               Main application pipeline
    config.py                       Centralized configuration
    mapper.py                       Feature encoding
    decision_engine.py              Rule-based expert engine
    predict_action.py               Calibrated ML predictor
    ensemble_predict.py             PyTorch vision ensemble
    predict_pest.py                 Keras pest predictor (standalone)
    database_manager.py             SQLite persistence
    generate_synthetic_data.py      Synthetic data generator
    train_tabular.py                XGBoost training pipeline
    train_cv.py                     5-Fold cross-validation
    calibrate.py                    Isotonic calibration pipeline
    train_pest_cnn.py              Basic Keras CNN training
    train_pest_cv.py               EfficientNet fine-tuning (Keras)
    view_logs.py                    Database consultation viewer
    utils.py                        Shared utility functions
  models/                           Serialized model artifacts
    ensemble_pest_model.pth         PyTorch triple-CNN checkpoint (~75 MB)
    pest_model_final.h5             Keras EfficientNet model (~76 MB)
    model_calibrated.pkl            Calibrated XGBoost model
    calibration_plot.png            Reliability diagram
  data/                             Datasets and database
    logs.csv                        Training observation logs
    agri_sense.db                   SQLite consultation archive
  reports/                          Exported digital prescriptions
  test_images/                      Sample leaf images for testing
  logs/                             Application runtime logs
  requirements.txt                  Python dependency manifest
  README.md                         Project documentation
```

8. Results and Observations

8.1 Tabular Classification Performance

The calibrated XGBoost classifier achieved a test-set classification accuracy of 90% on the structured observation log dataset, with a multiclass Brier score of 0.1557 indicating well-calibrated probability estimates. The reliability diagram confirmed close alignment between predicted confidence values and observed accuracy across ten calibration bins.

8.2 Vision Ensemble Inference

The PyTorch triple-CNN ensemble successfully performed leaf disease classification on test images, producing disease category predictions with associated probability scores. The ensemble correctly identified disease categories from the thirty-eight-class PlantVillage

taxonomy and the weighted fusion strategy (40/30/30) produced stable probability distributions across backbone architectures.

8.3 End-to-End System Validation

Complete end-to-end pipeline execution was validated with real-time Open-Meteo weather data retrieval for the city of Indore, India (resolved coordinates: latitude 22.72, longitude 75.86). The system successfully retrieved live atmospheric measurements (temperature: 25.5°C, wind speed: 4.0 m/s, sky condition: Cloudy), applied automatic weather parameter configuration, executed all three analytical pathways (rule engine, calibrated ML, vision ensemble), computed the hybrid CCI score (0.73), generated a complete advisory card with expert pesticide prescription, archived the consultation record in the SQLite database, and exported a digital receipt file.

9. Conclusion and Future Scope

The Smart Agriculture Expert System demonstrates a practical implementation of hybrid artificial intelligence for agricultural decision support. By combining transparent rule-based safety constraints with calibrated machine learning classification and multi-architecture deep learning disease detection, the system provides farmers with actionable, confidence-weighted recommendations grounded in both domain expertise and statistical evidence.

Future Enhancements

- **Mobile Application Interface:** Developing an Android-based front-end to enable field-level usage with smartphone camera integration for real-time leaf image capture.
- **Regional Language Support:** Extending advisory outputs to local Indian languages (Hindi, Marathi, Gujarati) to improve accessibility for non-English-speaking farming communities.
- **IoT Sensor Integration:** Incorporating direct readings from soil moisture sensors, humidity probes, and weather stations deployed at the field level.
- **Expanded Disease Taxonomy:** Training the vision ensemble on broader and more region-specific crop disease datasets beyond the PlantVillage benchmark.

- **Cloud Deployment:** Deploying the advisory pipeline as a REST API service for scalable, multi-user concurrent access.

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